MUHAMMAD IKRAM

ikram2500@gmail.com | +92 305 8876647 | linkedin.com/in/muhammad-ikram-088470190 | github.com/ikram2500

PROFESSIONAL SUMMARY:

- Data Science professional with hands-on experience building intelligent AI systems, multi-agent architectures, and end-to-end machine learning pipelines. Proficient in Python, SQL, and modern LLM frameworks including LangChain, AutoGen, and Pydantic. IBM-certified Data Science practitioner with demonstrated ability to design, develop, and deploy production-grade AI applications from concept to deployment.

TECHNICAL SKILLS

Programming & Data: Python, SQL, Pandas, NumPy, Scikit-learn, Matplotlib, Plotly, Dash, Folium

AI & LLM Frameworks: LangChain, LangGraph, LangSmith, AutoGen, Haystack, Pydantic, Agentic AI, A2A Protocol, MCP (Model Context Protocol)

Machine Learning & Deep Learning: Supervised & Unsupervised Learning, Feature Engineering, Model Evaluation, PyTorch, Neural Networks

APIs & Backend: FastAPI, REST API Design, OpenAI API, Groq API, OpenRouter API

Data Visualization: Matplotlib, Plotly, Dash, Folium, Looker Studio, Dashboard Development

Tools & Platforms: Git, GitHub, Streamlit, Jupyter Notebook, VS Code, Virtual Environments

CERTIFICATIONS

IBM Data Science Professional Certificate| 12-Course Specialization covering Python, SQL, Data Analysis, Machine Learning, and Capstone Project

PROJECTS

AI-Powered Academic Literature Review System

AutoGen | GPT-4o | Streamlit | arXiv API | Python

- Architected a multi-agent AI system using Microsoft AutoGen with specialized agents for arXiv paper retrieval and intelligent synthesis, automating end-to-end academic literature reviews with structured markdown report generation.
- Built an interactive Streamlit web interface enabling users to input research topics, configure retrieval parameters (1-10 papers), and export publication-ready literature review reports with paper summaries, key contributions, and actionable takeaways.
- Reduced literature review preparation time from hours to minutes by automating paper discovery, relevance filtering, and synthesis into structured markdown reports with linked PDFs, author metadata, and problem analysis.
- Designed an extensible agent architecture supporting future integration with Semantic Scholar, PubMed, and Google Scholar for multi-source academic research automation.

Data Analysis & Visualization Dashboard

Python | Pandas | Plotly | Dash | Looker Studio

- Performed end-to-end data analysis on complex datasets using Pandas and NumPy, implementing data cleaning, transformation, and statistical analysis workflows to extract actionable business insights.
- Developed interactive dashboards with Plotly, Dash, and Looker Studio, featuring dynamic filtering, real-time visualizations, and geographic mapping with Folium for comprehensive data storytelling.
- Applied machine learning techniques including regression, classification, and clustering using Scikit-learn to build predictive models and uncover patterns in structured data.

EDUCATION

M.Sc| The Islamia University of Bahawalpur | 2011

B.Sc. (Mathematics)| The Islamia University of Bahawalpur | 2006

ADDITIONAL INFORMATION

- Languages: English (Professional), Urdu (Native)
- Continuous learner actively exploring advancements in Agentic AI, LLM orchestration, and multi-agent systems through hands-on project development and online coursework.